

Electronic medical record privacy protection scheme based on attribute encryption technology

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Abstract—With the development of science and technology and the progress of medical and health system, electronic medical records with the advantages of large storage capacity and easy sharing have gradually replaced traditional paper medical records. In recent years, due to factors such as improper hospital management or hacker attacks, medical privacy leaks have occurred frequently, which not only caused serious privacy violations to patients, but also caused a negative impact on society that cannot be ignored. In order to prevent the leakage of medical privacy data during the sharing process and protect the electronic medical record data of patients, an access control scheme based on attribute encryption technology is proposed. The proposed scheme uses the ordered binary decision diagram (OBDD) as the access structure. Compared with the threshold access structure and the linear secret sharing access structure (LSSS), OBDD can represent any access strategy and support positive and negative values of attributes, and its expression efficiency is higher. The proposed scheme combines ciphertext attribute-based encryption (CP-ABE) with the national secret SM4 algorithm, and uses hybrid encryption to realize data protection of electronic medical records. After security analysis and experimental verification, the proposed scheme satisfies the selection plaintext security under the judgment bilinear Diffie-Hellman (DBDH) assumption, and has constant key length and high decryption efficiency, which has higher storage and computing performance advantages than other schemes.

Keywords—CP-ABE; OBDD; medical privacy protection; access control; SM4

I. INTRODUCTION

With the advancement of computer technology and the rapid development of the healthcare system, electronic medical records, as a modernized healthcare management method, are gradually replacing traditional paper-based medical records. Compared to paper records, electronic medical records offer advantages such as large storage capacity, convenient management, easy sharing, and quick retrieval of medical information. They not only improve the efficiency of diagnosis and treatment but also facilitate the efficient collection and analysis of medical data [1]. The widespread use of electronic medical records not only provides people with high-level, high-quality healthcare services but also offers convenient conditions for medical research. However, in today's society in the era of information "explosion," as the field of healthcare continues to develop and progress, the quantity of medical

data is rapidly increasing. While electronic medical record technology brings efficiency and convenience to people's lives, it also raises concerns about security issues, such as the leakage of private information.

As is well known, medical data contains rich value, encompassing not only basic information such as patient names, gender, and age but also sensitive information like patient phone numbers, bank accounts, residential addresses, and medical histories. In the eyes of malicious actors, medical data resembles a treasure trove, and the wealth it holds is enticing. According to the "2023 Healthcare Industry Data Security Report" released by Protenus, a healthcare analytics company in the United States, there were a total of 956 publicly disclosed medical data breach incidents in 2022, resulting in the exposure of the private health data of over 50 million patients. The leakage of medical privacy data not only infringes upon the privacy of patients but can also lead to lasting psychological harm, especially when sensitive personal information is exposed. Therefore, the protection of medical privacy data has become increasingly important and cannot be ignored.

The pathways for medical privacy breaches related to electronic medical records primarily include internal leakage within hospitals and external leakage from medical centers [1], typically occurring during the process of medical data release or sharing. To prevent the leakage of electronic medical record data, methods such as data encryption or the establishment of access control mechanisms can be employed to reduce the cost of leakage and enhance security. Ciphertext Policy Attribute-based Encryption (CP-ABE) [2][3] is a technique that achieves data encryption based on user attributes and provides a many-to-one access control mechanism. In CP-ABE, the encryptor formulates plaintext access policies, and the decryptor can only decrypt the plaintext if the access policy is satisfied. Therefore, CP-ABE technology is one of the methods for secure sharing of electronic medical record data.

However, as electronic medical record information continues to grow, the requirements for the storage and computational performance of the solution also increase. Traditional CP-ABE schemes [2] typically represent access policies in the form of access trees, requiring a significant number of bilinear operations in the implementation process. Moreover, both the key length

and ciphertext length increase with the number of attributes, leading to poor computational and storage performance. This makes traditional CP-ABE schemes unsuitable for the application scenarios of electronic medical records.

Therefore, this paper makes two main contributions. Firstly, in the process of attribute-based encryption, we employ Ordered Binary Decision Diagrams (OBDD) [4] as the structure for policy access, replacing the traditional access tree structure. The access structure of OBDD supports positive and negative values of attributes, enabling an efficient and flexible expression of access policies. Moreover, attribute-based encryption based on OBDD exhibits advantages such as short and constant decryption time compared to traditional attribute-based encryption. Secondly, we introduce the Chinese cryptographic algorithm SM4 [5], implementing a hybrid encryption of CP-ABE and SM4. This results in the design of a privacy protection scheme for electronic medical records based on attribute encryption. This scheme not only provides attribute-based access control but also addresses the high computational overhead of traditional CP-ABE schemes. It effectively reduces both computational and storage costs of the scheme.

II. RELATED RESEARCH

Attribute-Based Encryption (ABE) [6] integrates data encryption with access control policies and is categorized into two main types: Ciphertext Policy Attribute-Based Encryption (CP-ABE) [2~3] and Key Policy Attribute-Based Encryption (KP-ABE) [7]. Bethencourt and Waters [2] initially proposed a tree-based CP-ABE construction where data owners formulate access policies converted into an access tree structure. Data access is granted if the identity attributes of data consumers satisfy the access tree structure, allowing them to decrypt the plaintext using the private key. Cheung [8] introduced an AND-gate-based construction, relying on wildcard expressions to express access policies. Waters [9] proposed a construction based on Linear Secret Sharing Scheme (LSSS) structures, using the Lewko-Waters [10] algorithm to transform the access tree into an LSSS matrix, providing more flexibility in expressing access policies but still lacking support for logical NOT operations.

However, these approaches have limitations in expressing complex access policies efficiently. The algorithm complexity is often correlated with the number of attributes [11], resulting in high computational and storage costs. Furthermore, embedding access policies in ciphertext may lead to the leakage of sensitive information about data owners and data consumers [12]. Consequently, several scholars have proposed targeted solutions to address these challenges.

Lai[13] proposed a policy hiding construction scheme based on Linear Secret Sharing Scheme (LSSS), but this scheme can only conceal partial access policies. Phuong[14] introduced a policy hiding construction scheme based on AND gates, with a constant ciphertext

size and achieving Chosen Plaintext Attack (CPA) security. Li et al.[15] presented an outsourced computation scheme for CP-ABE, introducing virtual attributes in access policies, leading to the creation of a new policy tree. Most of the key generation and decryption computations are delegated to a more powerful cloud server, enhancing the computational efficiency of the scheme. Li[16] put forth a scheme based on Ordered Binary Decision Diagram (OBDD) access structures, leveraging the unique structure of OBDD for efficient expression of access policies. This approach exhibits characteristics such as constant key size and high decryption efficiency. However, its performance is closely tied to the specific OBDD structure, especially when there are too many effective paths in the OBDD, leading to encryption overhead surpassing other CP-ABE schemes.

Building upon the insights from the aforementioned schemes, this paper proposes a hybrid encryption scheme based on attribute encryption technology, combining CP-ABE algorithm and SM4 algorithm. Using OBDD as the access structure allows flexible expression of access policies through Boolean functions, supporting the occurrence of a single attribute multiple times in the access policy. This approach ensures the security of electronic medical record data through access control and data encryption, and compared to other schemes, it exhibits lower computational and storage costs in key generation and decryption computations.

III. PRELIMINARY KNOWLEDGE

This section serves as a preparatory overview for the subsequent chapters, providing introductions to the CP-ABE framework structure, the Decisional Bilinear Diffie-Hellman assumption [17] (DBDH), Ordered Binary Decision Diagram structure [4] (OBDD), and the SM4 algorithm. The proposed scheme is designed by integrating elements from the CP-ABE framework, OBDD access structure, SM4 algorithm, and other relevant components.

A. Prime Order Bilinear Group

Assuming G_0 and G_1 are two cyclic multiplicative groups of order P , and P is a prime number, the bilinear mapping $e: G_0 \times G_0 \rightarrow G_1$ satisfies the following properties:

(1) Bilinearity: For any $\forall u, v \in G_0, \forall a, b \in \mathbb{Z}_p$, there exists $e(u^a, v^b) = e(u, v)^{ab}$.

(2) Non-Degeneracy: $\exists c \in G_0, e(c, c) \neq 1$

(3) Computability: $\forall u, v \in G_0, e(u, v)$ is computable for all relevant instances.

B. CP-ABE Basic Framework

The basic process of CP-ABE mainly includes the following four steps [20]: initialization algorithm, key

generation algorithm, encryption algorithm, and decryption algorithm.

C. Decisional Bilinear Diffie-Hellman (DBDH) assumption

Assuming e is a bilinear map $e: G_0 \times G_0 \rightarrow G_1$ and randomly selecting $a, b, c, d \in_{\mathbb{R}} Z_p^*$, where g is a generator of the bilinear group G_0 . The DBDH assumption states that, in polynomial time, no adversary can distinguish between two tuples (g^a, g^b, g^c, g^{abc}) and (g^a, g^b, g^c, g^z) . The advantage of the adversary is defined as:

$$\text{Adv}_{G_0}^{\text{DBDH}}(\lambda) = |\Pr[\text{Adv}(g^a, g^b, g^c, g^{abc}) = 1] - \Pr[\text{Adv}(g^a, g^b, g^c, g^z) = 1]| \quad (6)$$

If $\text{Adv}_{G_0}^{\text{DBDH}}(\lambda)$ is non-negligible, the DBDH hypothesis is considered to be true.

D. Ordered Binary Decision Diagram

An Ordered Binary Decision Diagram (OBDD) is an acyclic directed graph structure that efficiently represents Boolean functions. It can be utilized to express access policies.

An OBDD consists of two types of nodes: decision nodes and terminal nodes. For decision nodes, they have two child nodes. When the current decision node takes a value of 1, its child node is called the high node; when the value is 0, the child node is called the low node. Terminal nodes have values of 0 or 1. In the OBDD structure, for each path from the root node to a terminal node, the order of variable occurrences is the same, and the node attributes along each path must be unique. When a path starts from the root node (root) and reaches terminal node 1, it is referred to as a valid path.

E. OBDD Access Structure

This section describes how to represent an access policy using the OBDD structure. For an access policy P , the first step is to convert it into a Boolean function $f(x_0, x_1, \dots, x_{n-1})$. Assuming the system's attribute set is $i_0, i_1, \dots, i_{n-1}$, if a specific access policy requires possession of attribute i , the corresponding attribute i has a positive value; otherwise, the value is negative. Then, the Boolean function is transformed into an OBDD structure. The conversion process is as follows:

Assuming policy $P = x_0 \vee (x_1 \wedge x_2) \vee (x_2 \wedge x_3)$, first, convert it into a Boolean function $f(x_0, x_1, x_2, x_3) = x_0 + x_1 x_2 + x_2 x_3$. Then, using variables as input, calculate the values of f under different inputs and transform them into a binary decision tree, as shown in Figure 1. Finally, use reduction to convert the binary

decision tree into an ordered binary decision diagram (OBDD), as illustrated in Figure 2.

Here, x_0, x_1, x_2, x_3 represents an attribute, dashed lines represent branches leading to 0 nodes, solid lines represent branches leading to 1 nodes. All nodes in the OBDD are numbered, and the nodes in the figure can be represented by $\text{OBDD} = \{\text{Node}_{id}^i \mid id \in ID, i \in I\}$. ID represents the set of decision nodes, and I represents the complete attribute set of the OBDD access structure.

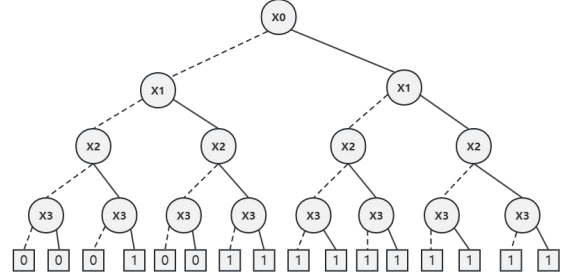


Fig. 1. Binary decision tree for $f(x_0, x_1, x_2, x_3)$

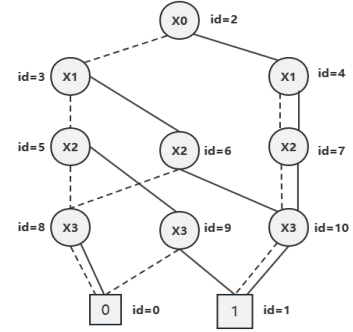


Fig. 2.

Fig. 2. OBDD structure of $f(x_0, x_1, x_2, x_3)$

Node_{id}^i can be represented as a quadruple $\langle i, id, low, high \rangle$, where i represents the node attribute, id is the node number, $high$ represents the number of the high node of the current node, and low represents the number of the low node of the current node. Terminal nodes have two states: 0 and 1, represented by squares labeled 0 or 1.

If a user's attribute set can form a valid path in the OBDD, then the user's attribute set is referred to as an authorized attribute set; otherwise, it is considered an unauthorized set.

F. SM4 Symmetric Encryption Algorithm

The main steps of the algorithm include:

- (1) Initialization Algorithm SetUp_{sm4}

- (2) Key Expansion Algorithm $KeyGen_{sm4}$
- (3) Encryption Algorithm $Encrypt_{sm4}$
- (4) Decryption Algorithm $Decrypt_{sm4}$

IV. SYSTEM SOLUTION COMPOSITION

This section discusses the system composition, security model, specific implementation and solution examples of the proposed solution.

A. System Components

The proposed solution in this paper is primarily applied to secure data sharing of electronic medical records between data owners and data users. The roles involved in the system include the Certification Authority (CA), Cloud Server (CS), Data Owner (DO), and Data User (DU). The specific model is illustrated in Figure 3:

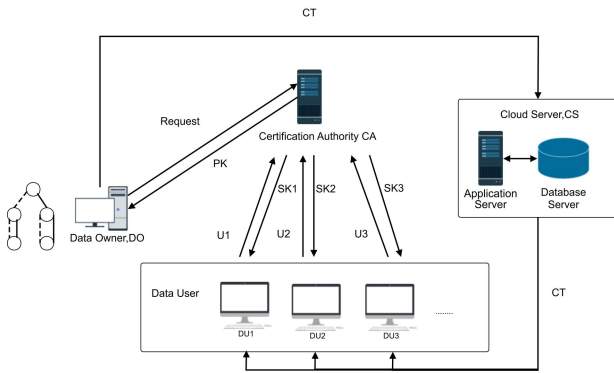


Fig. 3. System composition

(1) Certification Authority (CA): In this solution, the Certification Authority is considered trustworthy and is the core of the system. It is responsible for initializing the system, generating public parameters, and generating keys for users.

(2) Cloud Server (CS): Responsible for providing storage services for user-encrypted electronic medical record data. In this solution, the cloud server is considered a semi-trusted server, strictly executing mutually agreed-upon protocols but unable to guarantee data leakage.

(3) Data Owner (DO): Has the highest authority over medical record data, representing the patient. In this solution, the data owner can customize access policies, encrypt medical record data based on the specified access policies, and upload the encrypted data to the cloud server.

(4) Data User (DU): Represents healthcare professionals such as attending physicians and ward nurses. Requests the user's private key from the CA, and the CA provides the corresponding user's private key based on the attributes of the data user. Requests electronic medical record data from the cloud server; if the data user's attribute set satisfies the access policy of the ciphertext, the data user can decrypt the ciphertext using their private key to obtain plaintext data; otherwise, access is denied.

Given the high frequency and large demand for accessing electronic medical records in the current research context, there is a high efficiency requirement for data encryption and decryption. Therefore, this solution uses a hybrid encryption approach, combining an OBDD-based CP-ABE algorithm with the SM4 algorithm for encrypting electronic medical record data.

B. Security model

To demonstrate the security of the proposed solution, the scheme adopts the chosen plaintext security model based on the Chosen Plaintext Attack (CPA). This is elucidated through a game between the challenger and the adversary. The Chosen Plaintext Attack (CPA) can be described as follows: the adversary has the ability to choose any plaintext M and send it to the challenger, who processes it and returns the corresponding ciphertext CT to the adversary. The adversary, through analyzing the relationship between the plaintext M and the ciphertext CT , aims to gain algorithm-related information. This security challenge game mainly includes the following steps:

Initialization: The adversary initiates the game by selecting an access structure $OBDD_a$ and sends it to the challenger. The challenger initializes the system, generates relevant parameters, the system public key PK , and the system master key MK . The challenger then sends the parameters and PK to the adversary.

Query Phase 1: The adversary can choose different attribute sets U to make requests to the challenger and obtain the corresponding private key. In this phase, the requirement is that the attribute set U does not satisfy the $OBDD_a$. This phase can be repeated.

Challenge Phase: The adversary randomly selects two plaintext messages M_0, M_1 of equal length and sends them to the challenger. Upon receiving the plaintext, the challenger randomly chooses $x \in \{0, 1\}$ with a coin flip and encrypts M_x to obtain CT_x , finally sending CT_x to the adversary.

Query Phase 2, similar to Query Phase 1.

Guessing stage: After Adversary receives CT_x , it gives a guess x' for x , then the advantage of Adversary in winning the above game is defined as $Adv_{G_0}^{DBDH}(\lambda) = |\Pr[x' = x] - 1/2|$.

If the advantage of the adversary winning the game can be negligibly ignored in polynomial time, then the scheme is considered CPA-secure.

C. Specific implementation

This section presents the specific implementation of the electronic medical record privacy protection scheme based on attribute-based encryption (ABE) technology.

Due to the substantial computational overhead involved in bilinear pairings in the CP-ABE algorithm, the efficiency of encryption and decryption in CP-ABE is relatively low. In the context of this study, which focuses on secure sharing of electronic medical record data in healthcare systems where the access frequency is high, and the volume of access requests is significant, there is a need for efficient encryption and decryption algorithms.

To address this, the paper opts for a symmetric encryption algorithm to encrypt electronic medical record data. Subsequently, a CP-ABE scheme based on Ordered Binary Decision Diagrams (OBDD) is utilized to encrypt the symmetric key. By employing the CP-ABE scheme, access control to the symmetric key is achieved, facilitating access control for electronic medical record data. This approach aims to meet the high efficiency requirements for encryption and decryption algorithms in the context of frequent and substantial access to electronic medical records. This article's CP-ABE scheme supports positive and negative attributes. Suppose the system's attribute set N contains n elements, with attribute index

$\{0,1,2,3...n-1\}$. For each attribute i in N , if the attribute has a positive value, it is represented as $+i$, and if it has a negative value, it is represented as $-i$. The proposed scheme includes the following five algorithms.

(1) *SetUp*: Initialization Algorithm. This algorithm is executed by the Certification Authority (CA). The CA takes the security parameter λ as input and generates two cyclic groups G_0 and G_1 of order P , with a bilinear mapping $e: G_0 \times G_0 \rightarrow G_1$. The generator g belongs to the cyclic group G_0 .

A random selection $y, t_0...t_{n-1}, t'_0...t'_{n-1} \in Z_p$ is made, and the system's master key is set as

$$MK = (y, (t_i, t'_i) | i \in N) \quad (2)$$

The system public key is set to:

$$PK = (e, g, Y = e(g, g)^y, (T_i = g^{t_i}, T'_i = g^{t'_i}) | i \in N) \quad (3)$$

Among them, T_i and T'_i correspond to the positive and negative values of the attribute respectively.

(2) *KeyGen*(U, MK): Key Generation Algorithm. This algorithm is executed by the Certification Authority (CA). The CA generates the user's private key based on the user-provided attribute set U and the master key MK . Assume that the attribute set of the access policy is I , and $attr_i$ corresponds to attribute i , which is used to

store the key of attribute i . For $\forall i \in I$, the value of $attr_i$ is as follows:

$$attr_i = \begin{cases} t_i, i \in U \wedge i = +i \\ t'_i, i \notin U \vee i \in U \wedge i = -i \end{cases} \quad (4)$$

Select $r \in_R Z_p$, then the user's private key is:

$$SK = (D = g^{(r/\sum_{i \in I} attr_i)}, \hat{D} = g^{y-r}) \quad (5)$$

(3) *Encrypt*($PK, OBDD, M$): Encryption Algorithm. This algorithm is executed by the Data Owner (DO). Considering the relatively high time cost of the OBDD-based CP-ABE scheme during encryption, this scheme has been improved to reduce time overhead to some extent. Suppose there are three segments of electronic medical record information to be encrypted, denoted as M_1, M_2, M_3 , with the access policy set as

$$P_1 = P_a, P_2 = P_a \wedge P_b, P_3 = P_a \wedge P_b \wedge P_c \quad (6)$$

where P_a, P_b, P_c represents a sub-access policy. The DO randomly selects three symmetric keys of 128 bits each, denoted as K_1, K_2, K_3 , and encrypts them sequentially using the SM4 encryption algorithm, resulting in the ciphertext

$$CT_{M_x} = \text{Encrypt}_{sm4}(K_i, M_i), i = 1, 2, 3 \quad (7)$$

Then, the DO transforms P_a, P_b, P_c into an OBDD structure, denoted as

$$OBDD_x = \{Node_{id}^i | id \in ID, i \in I, x = a, b, c\}.$$

Step 1: For K_1 encryption, it is assumed that the number of effective paths of $OBDD_a$ is v , and the set of effective paths is represented as $P = \{P_1, P_2, P_3...P_v\}$. The encryption process is as follows: randomly select the secret value $s \in Z_p$ and calculate $\tilde{C} = K_1 \cdot Y^s, \hat{C} = g^s$,

For each valid path, we calculate

$$C_{P_j} = (\prod_{i \in I} T_i^*)^s = g^{(\sum_{i \in I} attr_i \cdot s)}$$

Among them, T_i^* corresponds to the value of attribute i , and $\sum attr_i$ is the sum of attributes on a single path. get ciphertext

$$CT_{K_1} = (OBDD_a, \tilde{C}, \hat{C}, \{C_{P_j} | P_j \in P\}) \quad (8)$$

Step 2: To encrypt K_2 , first use the SM4 encryption algorithm to encrypt K_2 using the key K_1 to obtain the encryption result.

$$CT' = \text{Encrypt}_{sm4}(K_2, K_1) \quad (9)$$

Then follow step 1 to use obdd to encrypt CT' again to obtain CT_{K_2} .

Step 3: To encrypt K_3 , first use the SM4 encryption algorithm, enter the key K_2 encrypt K_3 , and obtain the encryption result.

$$CT'' = \text{Encrypt}_{sm4}(K_3, K_2) \quad (10)$$

Then follow step 1 to use $OBDD_c$ to encrypt CT'' again to obtain CT_{K_3} . Finally got the ciphertext

$$CT = \{CT_{K_x}, CT_{M_x}\}, x = 1, 2, 3 \quad (11)$$

In this process, the number of attributes of P_2 and P_3 is reduced by using hybrid encryption, and the number of effective paths is reduced, thereby improving encryption efficiency.

(4) $CheckPerm(OBDD, U)$: Permission determination algorithm. If you want to decrypt ciphertext data, you first need to determine whether the attribute set U of the data visitor DU meets the OBDD access structure formulated by the data owner DO. If the access structure is met, the decryption algorithm is executed, otherwise it is deemed to be decrypted without permission. The steps of the algorithm are as follows:

Step 1: First find the root node root of the OBDD structure, whose corresponding id value is 2, and define it as the current node.

Step 2: According to the $Node_{id}^i$ defined in Section 2.3, obtain the quadruple $\langle i, id, low, high \rangle$ of the current node and analyze the current node. There are four processing situations:

① The current node is terminal node 0, the algorithm is terminated, and the return value is 0.

② The current node is terminal node 1, terminate the algorithm, and return value 1.

③ The current node is the decision-making node. If $i \in U \wedge i = +i$, select branch node *high* as the current node and continue to step 2.

④ The current node is the decision node. If $i \in U \wedge i = -i$ or $i \notin U$, select the 0 branch node *low* as the current node and continue to step 2.

(5) $Decrypt(CT, SK)$: Decryption algorithm.

Step 1: If the return value of $CheckPerm(OBDD, U)$ is 1, that is, the attribute set of DU satisfies the $OBDD_a$ access structure, decrypt K_1 :

$$\begin{aligned} K_1 &= \tilde{C} / e(\hat{C}, \hat{D}) \cdot e(C_{P_j}, D) \\ &= K_1 \cdot Y^s / e(g, g)^{s \cdot (y-r)} \cdot e(g, g)^{s \cdot r} \\ &= K_1 \cdot e(g, g)^{s \cdot y} / e(g, g)^{s \cdot y} \end{aligned} \quad (12)$$

Step 2: If the return value of $CheckPerm(OBDD, U)$ is 1, that is, the attribute set of DU satisfies the $OBDD_a$ and $OBDD_b$ access structures. Decrypt K_2 in the same way as step 1 to get CT' , and then use K_1 as the key to decrypt using the SM4 algorithm to get:

$$K_2 = \text{Decrypt}_{sm4}(CT', K_1) \quad (13)$$

Step 3: If the return value of $CheckPerm(OBDD, U)$ is 1, indicating that the attribute set of Data User (DU) satisfies the access structure $OBDD_a$, $OBDD_b$, and $OBDD_c$, decrypt K_3 using the method described in Step 1 to obtain CT'' . Then, use K_2 as the key to decrypt with the SM4 algorithm, resulting in

$$K_3 = \text{Decrypt}_{sm4}(CT'', K_2) \quad (14)$$

Finally, the electronic medical record data is obtained through the SM4 decryption algorithm.

$$M_x = \text{Decrypt}_{sm4}(CT_{M_x}, K_x), x = 1, 2, 3 \quad (15)$$

D. Solution Example

This section will provide an example of the proposed scheme. As shown in Figure 4, a patient wants to add access control to their electronic medical records, allowing only neurologists, ward nurses, or pain management doctors to have permission to view the records. Any other individuals are considered unauthorized to access. Setting the access policy $P = attr_1 \vee attr_2 \vee attr_3$ and $attr_1, attr_2, attr_3$ respectively represents the attributes of neurologists, ward nurses, and pain management doctors, transforming them into boolean functions $F(x_0, x_1, x_2) = x_0 + x_1 + x_2$, and finally obtaining the access structure $OBDD_b$. Using a hybrid encryption approach with the SM4 encryption algorithm and the CP-ABE encryption algorithm, the electronic medical record data M is encrypted, resulting in ciphertext CT , which is uploaded to the cloud server CS. When a neurologist requests CT from CS and provides their attribute set U_1 as input, the permission determination algorithm is executed. The algorithm determines that U_1 satisfies the $OBDD_b$, and the

[illegible]

V. SECURITY ANALYSIS

threshold access structures and LSSS access structures. The specific comparison is shown in Table 1. For ease of description, we take the access policy P_t as an example, assuming its corresponding Boolean function is:

$$f_t(x_0, x_1, x_2, x_3) = x_0x_1' + x_1x_2' + x_2x_3' + x_3x_0' \quad (20)$$

Among them, x_0, x_1, x_2, x_3 corresponds to the positive value of the attribute, x_0', x_1', x_2', x_3' corresponds to the negative value of the attribute, and the OBDD structure, LSSS access structure, and threshold access structure corresponding to P_t are shown in Figure 5 and Figure 6.

TABLE I. EXPRESSION EFFICIENCY AND ABILITY COMPARISON

	Access Structure	AND	OR	NOT	Number of variables
Ref[19]	Tree	√	√	X	4
Ref[18]	LSSS	√	√	X	8
Our	OBDD	√	√	√	8

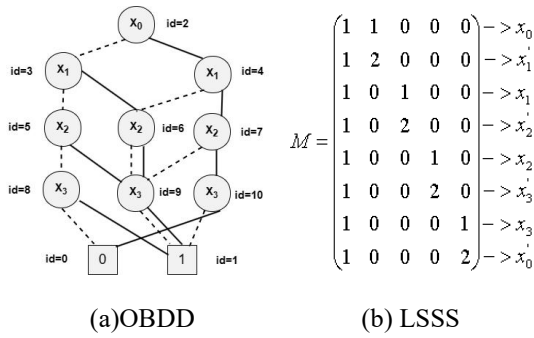


Fig. 5. OBDD and LSSS access structure of P_t

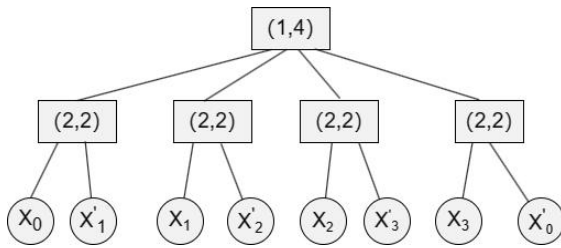


Fig. 6. Threshold access structure of P_t

Due to limitations in threshold access structures and LSSS access structures, which do not support the NOT operation, new variables are needed to separately represent negative values, disrupting the positive-negative relationships of variables. In threshold access structures, the number of variables is related to the number of leaf nodes. In LSSS access structures, the number of variables is related to the number of rows in the matrix. Therefore, in LSSS access structures and threshold access structures, the number of variables is associated with their expression efficiency. From Figure 5 and Figure 6, it can be observed

that the OBDD access structure requires 4 variables to represent the access policy, while LSSS access structures and threshold access structures require twice the number of variables. Therefore, OBDD access structures have a higher expression efficiency compared to the other two access structures, as shown in Table 1

In summary, the OBDD access structure employed in this paper demonstrates superior capabilities and efficiency in expressing access policies compared to common LSSS access structures and threshold access structures. For any given access policy, this paper utilizes Boolean operations to transform it into a Boolean function, subsequently representing the Boolean function using the OBDD access structure. This approach not only supports positive and negative attribute values but also allows for the repeated occurrence of a single attribute in the access policy, providing a more flexible expression method.

B. Theoretical Analysis

This section provides a performance comparison between the proposed attribute-based encryption scheme and other attribute-based encryption schemes [18~19]. The discussion primarily revolves around metrics such as access structure, user private key length, ciphertext length, private key generation cost, encryption cost, decryption complexity, and others. For clarity, Table 2 provides explanations of the relevant parameters. Tables 3 and 4 present specific metrics for evaluation.

TABLE II. PARAMETER DESCRIPTION

Parameter	Meaning
$ G_1 $	The size of a single element of G_0
$ G_1 $	The size of a single element of G_0
$ I $	Number of attributes in the access policy
$ U $	Number of user attributes
P	The finite number of paths of OBDD
T	Bilinear pairing operation time overhead
E_{G_0}	The time cost of a single exponentiation operation on G_0
E_{G_1}	The time cost of a single exponentiation operation on G_1
σ	Minimum number of attributes to decrypt
CT_m	Ciphertext obtained using SM4 encryption algorithm

TABLE III. STORAGE PERFORMANCE COMPARISON

	Access Structure	User private key length	Ciphertext length
Ref[19]	Tree	$(1 + 2 U) G_0 $	$ G_1 + (2 I + 1) G_0 $
Ref[18]	LSSS	$(2 + U) G_0 $	$ G_1 + (2 I + 1) G_0 $
Our	OBDD	$2 G_0 $	$ G_1 + (P + 1) G_0 + CT_m$

TABLE IV. COMPUTE PERFORMANCE COMPARISON

	Private key generation loss	Encryption loss	Decryption complexity
Ref[19]	$(1+3 U)E_{G_0}$	$E_{G_1} + (1+2 I)E_{G_0}$	$O(\sigma)$
Ref[18]	$(2+ U)E_{G_0}$	$E_{G_1} + (1+3 I)E_{G_0}$	$O(\sigma)$
Our	$2E_{G_0}$	$E_{G_1} + (1+P)E_{G_0}$	$O(1)$

As shown in Table 3 and Table 4, the user private key in this scheme has a fixed length, which is only the size of two G_0 group elements, and the key generation loss is $2E_{G_0}$, and the time complexity of the key generation algorithm is $O(1)$. During the decryption process, only when the user's attribute set satisfies one of the valid paths of OBDD, the corresponding C_{P_j} can be used to decrypt the SM4 encryption algorithm key, where the time overhead is $2E_{G_1} + 2T$. For the encryption and decryption efficiency of SM4, due to SM4 The steps of round function iteration, S-box replacement, P-box replacement, etc. are fixed, and their efficiency is related to the number and size of plaintext data blocks. Therefore, for fixed-length ciphertext decryption, the decryption time complexity of this scheme is $O(1)$. For encryption, the encryption time complexity and ciphertext length of this scheme depend on the number of effective OBDD paths P . The other two schemes depend on the number of attributes in the access policy. Therefore, when P is small, this scheme The encryption efficiency and ciphertext storage performance will be significantly improved.

In summary, this scheme is superior to the other two schemes in terms of key generation efficiency, key storage performance, and decryption efficiency.

C. Experiment Analysis

This section evaluates the proposed scheme through simulation experiments. To visually present the evaluation results, we compare the proposed scheme with references [18~19] in two steps. The first experiment uses a single plaintext segment to assess the performance of the proposed scheme and references[18~19] in key generation, encryption, and decryption algorithms under a single access policy. The second experiment uses multiple plaintext segments to evaluate the performance of the proposed scheme and references [18~19] under multiple access policies.

The experimental environment for the proposed scheme is a laptop with an AMD Ryzen 7 5800H with Radeon Graphics (3.20 GHz) processor, 8GB memory, and Windows 11 x64. The scheme is based on the JPBC cryptographic library, using a 512-bit type A pairing-friendly elliptic curve to construct a 160-bit elliptic curve group for experiments. The results represent the average values of 30 runs. The results of the first experiment are

shown in Figures 7, 8, and 9, while the results of the second experiment are shown in Figures 10, 11, and 12.

Figure 7 illustrates the efficiency comparison of user key generation under a single access policy. Through experimental analysis, as the number of user attributes increases, the key generation time in references [18~19] shows a linear growth trend. Reference [18], due to the use of hash operations in the key generation algorithm, incurs more time than reference [19]. In contrast, the proposed scheme's key generation is unaffected by the number of attributes, with a time cost of approximately 13ms.

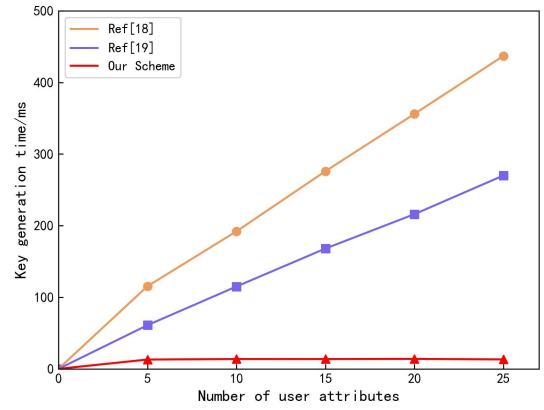


Fig. 7. Key generation loss under a single access policy

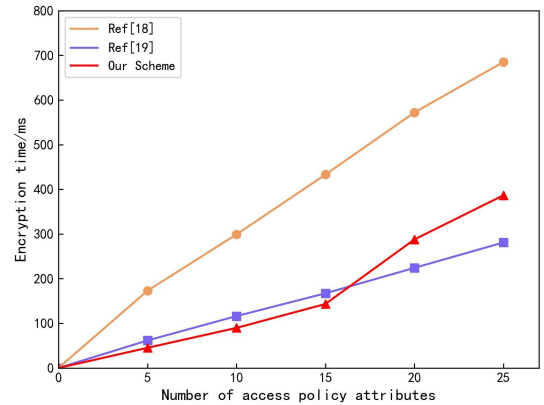


Fig. 8. Encryption time loss under a single access policy.

Figure 8 illustrates the efficiency comparison of encryption under a single access policy. When using the same policy, the encryption time cost in references [18~19] is positively correlated with the number of attributes in the access policy. The proposed scheme utilizes an OBDD access structure, and the encryption time cost is influenced by the number of valid paths in the OBDD. As the number of attributes in the access policy increases, the number of valid paths also increases, leading to a positive correlation between the encryption cost and the number of

attributes in the access policy. Figure 8 also shows that when the number of attributes is relatively small (less than 15), the encryption time cost of the proposed scheme is lower than that of references [18~19].

Figure 9 shows the comparison of decryption efficiency under single access strategy. The decryption overhead in literature [18~19] is positively correlated with the number of user attributes. The decryption overhead of this scheme is constant and the time is short, about 8ms. In terms of decryption, It is significantly better than the literature [18~19].

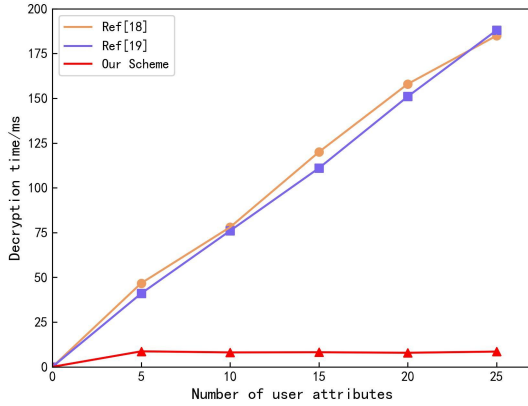


Fig. 9. Decryption time loss under a single access policy

In the second evaluation, this paper actually carried out experimental research using three access strategies as examples. Figure 10 shows the comparison of user private key generation efficiency under multiple access policies. After experimental analysis, under the multi-access strategy, the key generation time of the literature [18~19] still shows a linear increase with the number of user attributes, while the key generation time of this scheme is not affected by the number of user attributes, showing a stable situation. And compared to a single access policy,

In terms of key generation time overhead, as the number of attributes increases, the difference between this scheme and the literature [18~19] further increases, and the performance becomes even better.

Figure 11 shows the encryption efficiency comparison under multiple access policies. After experimental analysis, under the multi-access strategy, the encryption time loss of this scheme is the smallest among the three schemes. And compared with the single access strategy, through comparison with the literature [19], when the number of attributes is large (more than 15), the encryption time loss of this scheme is significantly improved.

Figure 12 shows the comparison of decryption efficiency under multiple access policies. After experimental analysis, it is found that the decryption overhead of this scheme is still constant under the multi-access strategy, and the time overhead is minimal. The

decryption time loss in literature [18~19] is close to that of the previous one, and is positively correlated with the number of attributes.

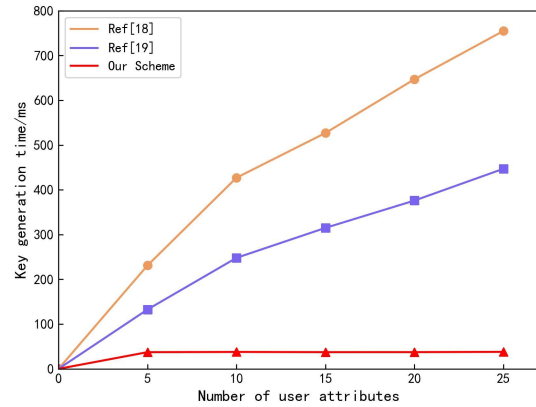


Fig. 10. Key generation loss under multiple access policies

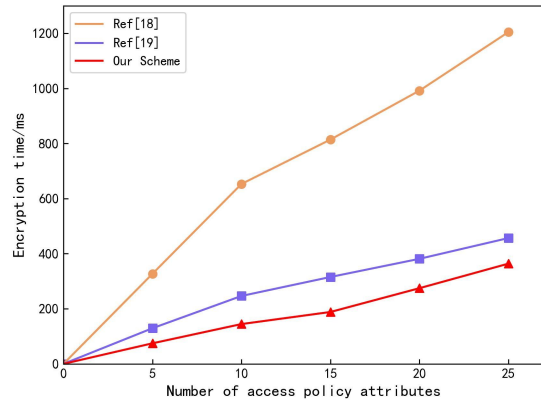


Fig. 11. Encryption time loss under multiple access policies

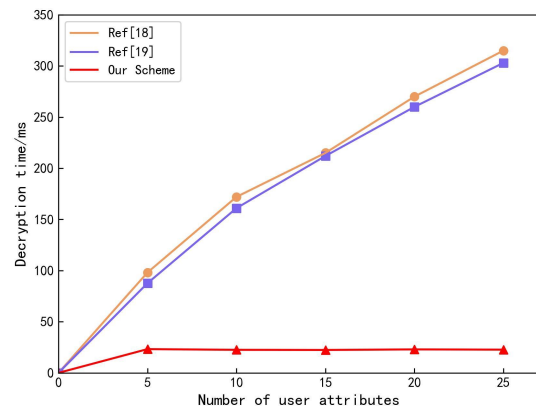


Fig. 12. Decryption time loss under multiple access policies

In summary, after experimental comparative analysis, whether under single access strategy or multiple access strategy, the time cost of key generation and decryption of this scheme is significantly better than that of the literature [18–19]. In terms of encryption overhead, the encryption overhead of this scheme is affected by the number of effective paths of OBDD. Under a single access policy, when the number of access policy attributes is small, this scheme has an advantage in the comparison of encryption overhead; and when the number of attributes is larger. In many cases, the encryption time overhead of this scheme will be higher than that in literature [19] due to differences in strategies (too many effective paths), which is improved in the multi-access strategy.

VII. CONCLUSION

In order to protect the data security of electronic medical records during the sharing and use process, this paper proposes a privacy protection scheme for electronic medical records based on attribute encryption technology. The proposed scheme combines CP-ABE with SM4 and uses hybrid encryption to achieve electronic privacy protection. Access control and data protection of medical records. This solution uses OBDD as the access structure, which can use Boolean operations to flexibly represent the access policy. It supports positive and negative attribute values, allows a single attribute to appear multiple times in the policy, and is superior to common threshold access structures and common threshold access structures in terms of expression efficiency and capabilities. LSSS access structure. When data is encrypted, the data owner formulates an access policy and converts it into an OBDD structure. When data is decrypted, only when the visitor's attribute set satisfies the OBDD structure can it be decrypted. The encryption overhead of the scheme is related to the effective path of OBDD. The key generation and decryption performance is higher than other schemes. It has the advantages of constant key and high decryption efficiency.

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